Conduct a comprehensive survey of public datasets for USV (Unmanned Surface Vehicle) obstacle detection and semantic segmentation. Focus specifically on: WaterScenes Dataset, MaSTr1325, MODD2, LaRS, USVInland, SMD, SPSCD Maritime Dataset, KOLOMVERSE, Pohang Canal Dataset.

For each dataset, provide:

1. **Data Format:** Resolution, FPS, file structure.
2. **Annotation Types:** Semantic masks (PNG), bounding boxes (YOLO/COCO), or panoptic segmentation.
3. **Class Taxonomy:** List the specific classes used (e.g., distinction between 'static obstacle', 'dynamic vessel', 'water', 'sky').
4. **Collection Environment:** River vs. Sea, weather conditions, lighting variations.
5. **Sensor Setup:** What cameras/sensors were used to record them?

Comprehensive Survey of Public Datasets for USV Obstacle Detection and Semantic Segmentation

1. WaterScenes Dataset

Data Format

WaterScenes provides 1920×1080 HD resolution imagery with 4D radar point clouds. The camera operates at approximately 30 Hz (synchronized with 4D radar at 10-15 Hz). Data is organized into RGB images (JPG format), radar files (CSV format for point clouds), and temporal radar accumulations (3-frame and 5-frame windows). The dataset encompasses 54,120 synchronized multimodal frames with over 200,000 annotated objects.[1][2]

Annotation Types

The dataset offers multi-modal annotations across six perception tasks: (1) 2D bounding boxes in COCO format for camera images, (2) pixel-level semantic segmentation masks, (3) instance segmentation masks, (4) free-space segmentation masks, (5) waterline segmentation, and (6) 3D point-level annotations for radar point clouds with semantic labels.[1-1]

Class Taxonomy

Thing classes (8 dynamic/static obstacles): pier, buoy, sailor (person on vehicle), ship, boat, vessel, kayak, and no-object (radar-only designation). Stuff classes for segmentation tasks include free-space and waterline.[1-2]

Collection Environment

Collection occurred across diverse water environments including lakes, rivers, and coastal areas. Environmental diversity encompasses day/night variations, clear and adverse weather (rain, fog), sun glare conditions, and varying water turbulence. The dataset addresses multi-modal fusion challenges in diverse lighting and weather conditions.[1-3]

Sensor Setup

The USV is equipped with a 4D mmWave radar providing range, Doppler velocity, azimuth, and elevation information; a monocular RGB camera (1920×1080); GPS/GNSS for position and heading; and a 6-axis IMU for orientation and angular velocity measurements.[1-4]

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2. MaSTr1325

Data Format

The dataset contains 1,325 diverse images at 512×384 pixel resolution, captured over two years from a real coastal USV. Images are provided as PNG/JPG with corresponding per-pixel segmentation masks (PNG format). Data is synchronized with onboard GPS and IMU measurements.[3][4]

Annotation Types

All images feature per-pixel semantic segmentation masks. Annotation boundaries include "unknown" category labels to account for edge uncertainties between semantic regions. Pixel labels encode three main categories: obstacles/environment (0), water (1), sky (2), and ignore regions (4).[4-1][3-1]

Class Taxonomy

The taxonomy is simplified to three primary classes—water, sky, and obstacles—with obstacles treated as a generic category encompassing diverse types: cargo ships, sail boats, portable piers, swimmers, rowers, buoys, and seagulls. The emphasis on generic obstacle detection prioritizes navigation safety over fine-grained object categorization.[3-2][4-2]

Collection Environment

Data collection spanned two years in the Gulf of Koper, Slovenia, covering realistic coastal surveillance conditions. Extensive attention to weather diversity includes foggy conditions, partly cloudy with sunrise, overcast, and sunny scenarios. The dataset explicitly captures seasonal and temporal variations typical of coastal operations.[4-3][3-3]

Sensor Setup

The platform features a forward-facing monocular camera synchronized with GPS and 6-axis IMU. The integration of inertial measurements with visual data enables IMU-assisted semantic segmentation approaches.[3-4]

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3. MODD/MODD2

Data Format

MODD v1.0 consists of 12 fully-annotated marine videos extracted from USV operations. MODD v2.0 upgrades to stereo video with IMU integration. Frame rates typically 25-30 fps. Videos include pixel-level segmentation masks synchronized with sensor data.[5]

Annotation Types

Both versions provide per-pixel semantic segmentation masks. MODD v1.0 and v2.0 annotate three natural regions: sky, shore/land, and sea. MODD v2.0 includes stereo correspondence annotations enabling disparity/depth estimation and IMU orientation data for sensor fusion approaches.[5-1]

Class Taxonomy

Three semantic classes: sky (atmospheric regions), shore/land (static obstacles including banks, piers, structures), and sea/water. Obstacles are implicitly represented through the shore annotation, which separates static obstacles from navigable water regions.[5-2]

Collection Environment

Data capture occurred in diverse marine/coastal environments with varying weather conditions and visual challenges. The stereo formulation of MODD v2.0 provides baseline depth context. Common challenges include water reflections, glare, and horizon detection ambiguities.[5-3]

Sensor Setup

MODD v1.0 uses a monocular camera. MODD v2.0 integrates stereo cameras (for depth context), 6-axis IMU, and GPS. The stereo baseline and inertial data support multi-modal fusion

research for inland and coastal navigation.[5-4]

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4. LaRS (Lakes, Rivers, Seas)

Data Format

LaRS provides 4,000+ manually labeled keyframes with 9 preceding frames per keyframe (totaling 40,000 frames). Images are extracted from diverse sources representing lakes, rivers, and coastal seas. Per-frame panoptic annotations include 19 global scene attributes.[6][7]

Annotation Types

Panoptic segmentation combines instance segmentation (per-object masks with instance IDs) and semantic segmentation (per-pixel class labels). Bounding boxes annotate all dynamic obstacles. Scene-level attributes (19 global descriptors) characterize illumination, weather, environment type, and water conditions.[6-1]

Class Taxonomy

Thing classes (8 dynamic/static obstacles with instance IDs): ship/large vessel, boat/small vessel, person/swimmer, buoy, floating debris, jetski/personal watercraft, kayak, surfboard. Stuff classes (3 static semantic regions): water, sky, vegetation/shore. Scene attributes (19): foggy, rainy, windy, sunny, cloudy, night, hazy, glare, reflection-heavy, waves, urban, rural, port, open water, etc.[6-2]

Collection Environment

LaRS explicitly targets maximum diversity in maritime obstacle detection. Collection spans inland lakes, river navigation scenarios, and open coastal waters. Geographic diversity covers multiple continental locations with distinct environmental characteristics. Weather and lighting variations are comprehensively represented through the 19 scene attributes.[7-1][6-3]

Sensor Setup

Multi-source collection: manual USV capture, public video sources, and existing dataset incorporation. Primarily monocular RGB camera-based, enabling compatibility with single-camera USV systems while supporting panoptic segmentation research.[6-4]

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5. USVInland

Data Format

Multi-sensor dataset spanning 26+ km of inland waterway trajectories. Stereo camera provides 640×400 (low-res) or 1280×800 (high-res) resolution. Lidar operates at 10 Hz with 360° horizontal FoV. Radar produces point clouds. Synchronization achieves 25 ms maximum drift. Data structured as 27 continuous raw sequences from 100-2600 seconds duration.[8]

Annotation Types

Water segmentation benchmark includes 700 images with polygon annotations created via LabelMe toolbox. SLAM benchmark provides GPS/IMU ground truth (RTK Fixed mode only) for trajectory estimation. Stereo matching benchmark includes 324 image pairs with lidar-derived depth ground truth via ICP point cloud aggregation.[8-1]

Class Taxonomy

Water segmentation task: binary classes (water area vs. non-water/banks/obstacles). No fine-grained object detection classes; focus on navigation-critical water boundary identification.[8-2]

Collection Environment

Inland waterways in Chinese cities (Xi'an, Ningbo, Hangzhou) during May-August 2020. Comprehensive weather coverage: clear sun, strong glare, overcast, mist, rain, heavy fog. Waterway widths 10-40 m with lengths 100-3400 m. Challenges include GPS signal denial under bridges, vegetation occlusion, bankside structure reflections, and fog-induced lidar degradation.[8-3]

Sensor Setup

1× 16-beam 905nm Lidar (360°×32° FoV, 10 Hz, 100m range, 2cm range resolution). 1× Stereo RGB camera + IMU (640×400 or 1280×800, 20 Hz, 80mm baseline, global shutter, 95° HFoV). 3× Millimeter-wave radar (94° HFoV, 10 Hz, 18.08m max range). INS with 6-axis IMU (50 Hz) and 2× GPS RTK (5 Hz).[8-4]

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6. SMD (Singapore Maritime Dataset)

Data Format

Approximately 31,653 labeled frames from high-definition video (1080×1920 pixels, Canon 70D cameras). Captured in both Visual-Optical (VIS) and Near-Infrared (NIR) spectra. Video-sourced at 25-30 fps.[9]

Annotation Types

Bounding box object detection annotations (YOLO/COCO formats). Semi-automated instance segmentation label generation from bounding boxes using novel algorithmic approaches. Approximately 240,842 annotated objects across all frames.[9-1]

Class Taxonomy

10 object classes: ship/vessel, boat, buoy, ferry, speedboat, kayak, sailboat, flying bird/plane, other vessel, background/water (implied).[10][9-2]

Collection Environment

Singapore territorial waters with harbor and coastal conditions. Extensive environmental diversity: day/night lighting variations, clear to adverse weather conditions. Dual-spectrum capture (VIS and NIR) addresses challenges from strong glare, reflections, and low-visibility scenarios.[11][9-3]

Sensor Setup

Visual spectrum: Canon 70D camera (1080×1920 HD). Near-infrared thermal camera for adverse lighting conditions. Fixed surveillance platform capture from coastal/harbor vantage points.[11-1][9-4]

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7. SPSCD (Split Port Ship Classification Dataset)

Data Format

1,000+ images at 1920×1080 (HD resolution) extracted from video captured by Dahua LR surveillance camera. Collection spans February 2020 to December 2022. JPG image format with corresponding YOLO-format TXT annotation files.[12]

Annotation Types

Bounding box annotations in YOLO-Darknet format with normalized coordinates relative to image dimensions. One TXT file per image containing: class_id, center_x, center_y, width, height. Annotations capture port and starboard (both sides) vessel views.[12-1][3-5]

Class Taxonomy

12 maritime vessel categories curated with nautical engineering expertise:

1. Large Ferry, 2. Medium Ferry, 3. General Cargo Ship, 4. Container Ship, 5. Fishing Boat, 6. Passenger Ship, 7. RoRo Ship, 8. Sailing Yacht, 9. Motorboat, 10. Speed Craft, 11. Small Craft, 12. Pleasure Yacht

Classification considers visual similarity while respecting functional/operational distinctions between vessel types.[12-2]

Collection Environment

Port of Split, Mediterranean coast (Croatia). Video recordings from February 2020 to December 2022. Diverse times of day and year. Variable weather and sea state conditions. Focus on vessels entering/leaving port operations.[12-3]

Sensor Setup

Dahua LR long-range surveillance camera providing HD 1920×1080 video capability. Fixed land-based coastal monitoring platform with zoom and night vision capability.[12-4]

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8. KOLOMVERSE

Data Format

Approximately 2.151 million 4K resolution images (3840×2160 pixels) extracted from 5,845 hours of video. Source: 21 territorial waters across South Korea. Images in JPG format with YOLO format bounding box annotations.[13]

Annotation Types

Bounding boxes in YOLO format for object detection. Single-class annotations per object. Minimal annotation overhead optimized for large-scale dataset curation.[13-1]

Class Taxonomy

5 maritime object classes:

- Ship (large vessels)
- Buoy (navigation aids)
- Fishnet buoy (fishing-specific navigation)
- Lighthouse (stationary navigational structure)
- Wind farm (offshore installation)[13-2]

Collection Environment

South Korea's 21 territorial waters providing geographic diversity. Comprehensive environmental coverage: weather variations (clear, cloudy, fog, rain, snow), temporal variations (day, night, dusk/dawn), illumination changes, occlusion scenarios, variable viewpoints, wind speed variations (5-25 knots recorded), and visibility ranges. Represents one of the most comprehensive environmental coverage among maritime datasets.[13-3]

Sensor Setup

Multi-camera surveillance system deployed across Korean territorial waters. Long-range optical systems with night vision capability. 4K resolution capture optimized for small object detection in maritime scenarios.[13-4]

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9. Pohang Canal Dataset

Data Format

Multimodal dataset along 7.5 km restricted waterway route in Pohang, South Korea. Collection conducted under diverse weather and visual conditions. Data provided in ROS bag format with synchronized timestamp data. Resolution varies by sensor type.[14][15]

Annotation Types

SLAM baseline trajectory annotations using GPS/IMU data and visual odometry. Intrinsic and extrinsic sensor calibration parameters. Scene-level environmental classification (narrow canal, inner port, outer port, near-coastal). No pixel-level or instance segmentation annotations provided; focus on navigation and localization ground truth.[15-1][14-1]

Class Taxonomy

No explicit object detection classes. Environmental region types: water/navigable area, waterside obstacles, bank structures. Scene categorization: narrow canal (15-30m width), inner port (50-100m width, mixed docked vessels), outer port (100m+ width, large vessels), near-coastal (shallow water).[14-2][15-2]

Collection Environment

Restricted waters in Pohang, South Korea. Narrow canal area: 15-30m width, park and urban structures. Inner port: fish markets, shipyards, mixed small/large vessel activity. Outer port: passenger terminals, storage buildings, cruise/cargo ship traffic. Near-coastal: shallow water beyond harbor. Comprehensive weather documentation: clear conditions, rainy, foggy, nighttime. GPS signal denial in canal area under bridges.[15-3][14-3]

Sensor Setup

3× Lidar: 1× 64-channel (64-line), 2× 32-channel LiDAR systems (10-20 Hz)
1× Marine radar (solid-state, 10-15 Hz)
2× Visual RGB cameras (stereo configuration, ~30 Hz, wide baseline)
1× Infrared thermal camera
1× Omnidirectional camera (6 directional views)
GPS/RTK: 2 antennas with moving base support (5-10 Hz)
AHRS: 6-axis (50 Hz)
IMU: Integrated across platforms[14-4][15-4]

Comparative Analysis

Annotation Paradigm Coverage:
Datasets collectively span semantic segmentation (MaSTr1325, USVInland, MODD), panoptic segmentation (LaRS, WaterScenes), bounding box detection (SMD, SPSCD, KOLOMVERSE), and trajectory-based navigation annotation (Pohang Canal). This diversity supports development across multiple perception task families.[3-6][14-5][6-5][8-5][12-5][13-5]

Environmental Scope:
Coastal/marine focus (MaSTr1325, KOLOMVERSE, SPSCD) contrasts with inland waterway specialization (USVInland, Pohang Canal). LaRS uniquely addresses all three environments (lakes, rivers, seas), enabling cross-environment generalization research.[6-6][8-6][13-6]

Sensor Modality Progression:
Evolution from monocular RGB (MaSTr1325, MODD, LaRS) → stereo+auxiliary (USVInland) → advanced fusion (WaterScenes' 4D radar, Pohang Canal's 6-sensor suite). WaterScenes specifically addresses multi-modal robust perception under adverse conditions through 4D radar-camera fusion.[3-7][1-5]

Resolution and Detail:
KOLOMVERSE's 4K resolution (3840×2160) enables small object detection. Standard HD (1920×1080) prevalent in surveillance-sourced datasets (SMD, SPSCD). MaSTr1325 and MODD use lower resolutions (512×384, variable) reflecting embedded USV camera limitations. This spectrum supports diverse deployment scenarios.[8-7][12-6][13-7]

Obstacle Taxonomy Complexity:
Generic obstacle classes (MaSTr1325, USVInland, MODD) enable core navigation research. Fine-grained taxonomies (8-12 classes: LaRS, SMD, SPSCD) support specialized applications

(vessel classification, hazard differentiation). WaterScenes' 7-point taxonomy balances specificity with practical utility.[9-5][3-8][6-7][12-7][1-6]

Research Implications and Recommendations

Dataset Selection Criteria:

- **For real-time embedded systems:** MaSTr1325 (appropriate resolution), WaterScenes (multi-modal robustness)
- **For coastal harbor operations:** SPSCD (vessel-specific), KOLOMVERSE (comprehensive weather coverage)
- **For inland waterway navigation:** USVInland (river-specific challenges), Pohang Canal (multi-modal sensor integration)
- **For panoptic scene understanding:** LaRS (diversity), WaterScenes (multi-task)
- **For robust adverse-weather perception:** KOLOMVERSE, WaterScenes (4D radar), USVInland (fog/rain documentation)

Cross-Dataset Generalization:
The 9-dataset suite enables systematic generalization studies across environments (lakes→rivers→seas), sensor modalities (mono→stereo→4D fusion), and annotation paradigms (semantic→instance→panoptic). LaRS's source diversity (manual+public+existing datasets) enables controlled composition of cross-dataset training experiments.[6-8]

Emerging Research Directions:
The progression toward multi-modal perception (WaterScenes, Pohang Canal) reflects field maturation toward robust all-weather autonomous maritime operation. Limited panoptic annotation (only LaRS and WaterScenes) suggests opportunity for panoptic benchmark development. The absence of dataset-specific occlusion/reflection benchmarking despite widespread mention in literature indicates potential for specialized adversarial subset creation.[3-9][1-7][6-9]

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